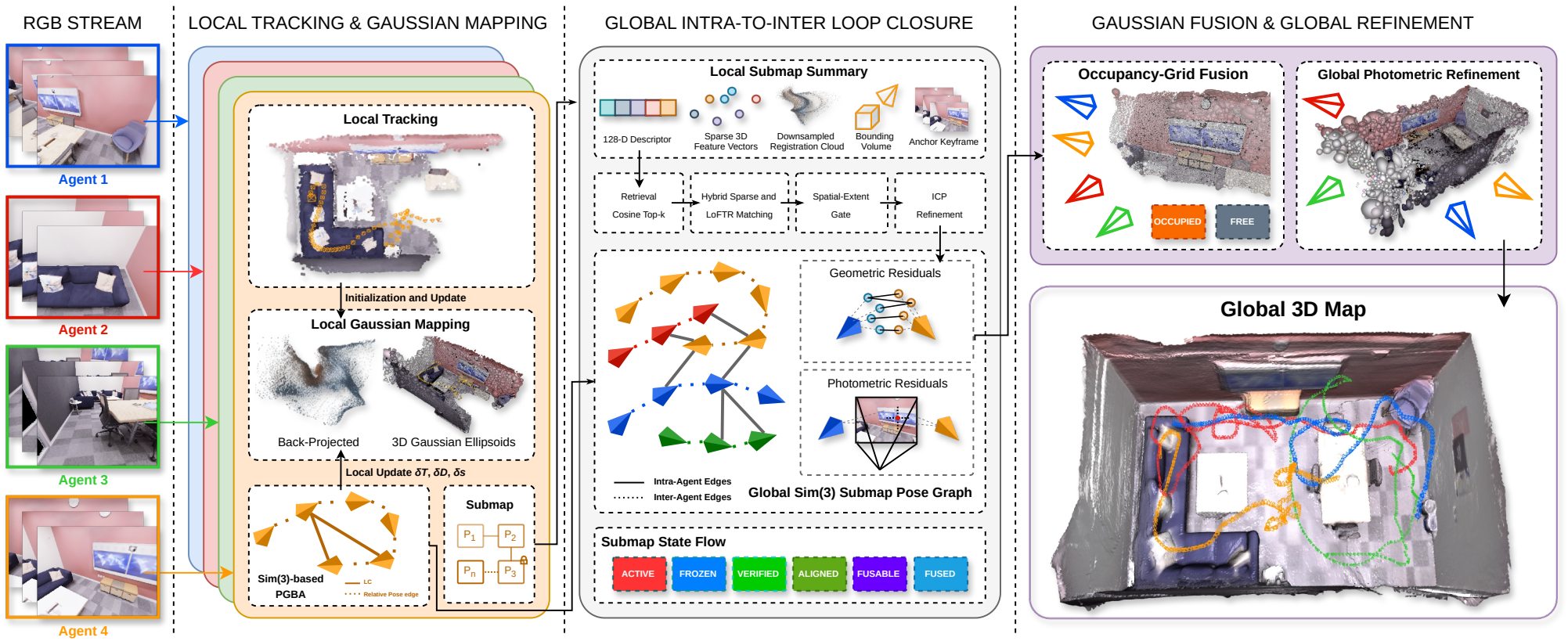

RGB STREAM
LOCAL TRACKING & GAUSSIAN MAPPING
GLOBAL INTRA-TO-INTER LOOP CLOSURE
GAUSSIAN FUSION & GLOBAL REFINEMENT
Agent 1
Agent 2
Agent 3
Agent 4
Local Tracking
Initialization and Update
Local Gaussian Mapping
Back-Projected
3D Gaussian Ellipsoids
Local Update δT, δD, δs
Sim(3)-based PGBA
LC
Relative Pose edge
Submap
P1
P2
Pn
P3
Local Submap Summary
128-D Descriptor
Sparse 3D Feature Vectors
Downsampled Registration Cloud
Bounding Volume
Anchor Keyframe
Retrieval Cosine Top-k
Hybrid Sparse and LoFTR Matching
Spatial-Extent Gate
ICP Refinement
Geometric Residuals
Photometric Residuals
Intra-Agent Edges
Inter-Agent Edges
Global Sim(3) Submap Pose Graph
Submap State Flow
ACTIVE
FROZEN
VERIFIED
ALIGNED
FUSABLE
FUSED
Occupancy-Grid Fusion
OCCUPIED
FREE
Global Photometric Refinement
Global 3D Map